

For the U-M ECT consultation

Things to share with the clinic, and concerns to raise before starting · 2026-05-20

For Dad, Mom, Leah, David,
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family team

What this document is for. A one-page set of bullets for Dad and Mom to bring into the U-M ECT consultation tomorrow. The top two boxes are the actionable summary — things to share with the U-M clinicians, and concerns to raise before starting ECT. The pages that follow develop the reasoning, the published evidence (including the more permissive Morikawa 2024 ECT-in-Lewy-spectrum data that Adam's earlier framing did not fully reflect), and a recommended question to put to the U-M ECT team. **The handout is not arguing against ECT in principle — it's arguing for completing the diagnostic workup that's already in motion before committing to it.**

THINGS TO SHARE WITH THE U-M ECT CLINIC

- **The March 2026 general anesthesia and the decline that followed.** Dad had general anesthesia for an Aquablation BPH procedure on 3/18/2026. Within ~1 week of recovery, he noticed depression returning. By 4/8 family observed new motor and expression changes. At 4/14 PCP visit, gait abnormality, decreased appetite, and unintentional weight loss were coded for the first time. Eight weeks later, he has not returned to his pre-procedure baseline. The temporal correlation is striking and consistent with documented post-anesthesia cognitive effects in older adults — though we cannot prove the anesthesia caused the decline.
- **Prodromal-synucleinopathy markers that predate the March GA.** Formal REM Sleep Behavior Disorder since 10/2017 (8.5 years). Balance problem coded 7/2025 — 9 months before the March procedure. Chronic constipation. Peripheral neuropathy on chart since 2018. Appetite-interoception dissociation (no spontaneous hunger registration) for 1–2 years. Per the Arnaldi 2021 phenoconversion scoring, this constellation places Dad in the highest-risk stratum for near-term progression to Parkinson's-spectrum or Lewy-body disease.
- **The pending U-M Movement Disorders evaluation includes DAT-SPECT.** Results will materially change ECT risk-benefit and monitoring plan — the published safety data for ECT in Lewy-spectrum disease (Morikawa 2024) comes from patients with confirmed abnormal DAT-SPECT, and a negative scan would substantially attenuate the prodromal-synucleinopathy concern. We are asking whether ECT can wait until this evaluation is complete.
- **Dad is a documented IV-ketamine responder.** February 2026: dramatic ~4-week response on duloxetine + brexpiprazole background. The Ann Arbor ketamine clinic has him scheduled to restart on June 8. He has not yet completed a full ketamine course at A2.
- **Documented activation-intolerance pattern** across six prior psychotropic agents in 2025 (aripiprazole, mirtazapine, brexpiprazole, olanzapine, pramipexole, lithium) — consistent with the neuroleptic-sensitivity pattern seen in Lewy-spectrum disease, though not specific to it.
- **Cardiac profile:** Left anterior fascicular block + sinus bradycardia (HR 49–53) + BP 112/58 — relevant for ECT-induced bradycardia and asystole risk.

CONCERNS TO RAISE BEFORE STARTING ECT

- **Defer until the U-M Movement Disorders evaluation is complete.** DAT-SPECT, structured motor exam, and formal cognitive baseline (MoCA at minimum) would each materially change the ECT decision, the technique selection, and the monitoring plan. The published literature does not mandate neurology clearance before ECT — but it also does not address what to do when the workup is already underway and the patient has a viable bridge treatment available.

- **The IV ketamine course at A2 should be the bridge during the workup interval.** Dad is a confirmed responder. Ketamine does not require general anesthesia. The June 8 restart is already scheduled. The published evidence (ELEKT-D, Anand 2023 NEJM) shows ketamine is non-inferior to ECT for nonpsychotic TRD with a meaningful cognitive-safety advantage (8.8-point delayed-recall advantage at end of treatment).
- **The cumulative general anesthesia exposure question.** An induction ECT course typically involves 6–12 brief GA exposures; maintenance ECT adds another 20–40 over a year. The published literature on cumulative GA exposure and cognitive decline in elderly adults (Schulte 2018 Mayo Clinic; Pennings 2025 MAAS cohort; Linassi 2022 OR 9.74 for executive function impairment) supports a dose-response relationship. ECT-anesthesia is briefer than surgical GA and uses TIVA, which lowers per-exposure risk — but the cumulative burden over a full induction + maintenance course is not negligible, particularly in a brain that has already shown a difficult post-GA recovery.
- **Acuity does not currently meet the published threshold for ECT urgency.** Standard indications for proceeding with ECT despite incomplete workup (per APA / Espinoza & Kellner 2022 NEJM / Unützer 2012 JAMA) are active suicidality, catatonia, psychotic features, food refusal, or life-threatening medical complications of depression. Dad's depression is severe (PHQ-9 = 24), but he is not currently in any of those scenarios. The Steffens 2024 NEJM review of TRD in older adults explicitly lists screening for coexisting cognitive impairment as part of TRD assessment — not a reason to skip evaluation.
- **If ECT does proceed, the technique should be optimized for the profile.** Right unilateral ultrabrief pulse (minimizes cognitive side effects and bradycardia risk). Propofol-based anesthesia (lower inflammatory profile, faster cognitive recovery vs barbiturates in elderly). Pre-ECT formal cognitive baseline (MoCA + Trail Making B) with serial monitoring every 3–4 sessions. Cardiology clearance with glycopyrrolate availability for ECT-induced bradycardia. Low threshold to discontinue if new motor or cognitive findings or delirium emerge.
- **The activation-intolerance pattern is itself information.** Multiple prior psychotropic agents poorly tolerated in 2025 — consistent with the neuroleptic sensitivity pattern seen in DLB / prodromal DLB. Worth flagging to the ECT team as a sensitivity signal that may extend to perioperative agents (propofol vs methohexital choice; succinylcholine considerations).

THE SINGLE QUESTION TO PUT TO THE U-M ECT TEAM

FOR DAD OR MOM TO ASK, IN THEIR OWN WORDS

Given that Dad's U-M Movement Disorders evaluation including DAT-SPECT is already underway, that he had a difficult post-anesthesia course after his March procedure, and that he is a documented IV-ketamine responder with an A2 restart scheduled June 8 — is it clinically reasonable to defer ECT until the neurology workup is complete and the ketamine course has been fairly tested? If you think ECT should proceed before then, what specific clinical findings or symptoms would justify that urgency?

What the published evidence supports about ECT in Dad's specific profile

The 5/20 OpenEvidence consultation (#19) on cumulative GA exposure risk in this patient was more nuanced than the family's earlier framing. It does not support a "do not do ECT" position. It supports a "complete the diagnostic workup first, use the right technique, complete the ketamine bridge before committing" position. The reasoning, condensed:

ECT IN LEWY-SPECTRUM DISEASE IS ACTUALLY SAFER THAN WE'D BEEN TREATING IT

- **Morikawa 2024** is the most directly relevant safety study: retrospective analysis of ECT in 25 patients with confirmed DLB (n=12) or prodromal DLB/MCI-LB (n=13), all with abnormal DAT-SPECT and MIBG scintigraphy. 96% showed at least minimal improvement; 60% showed remarkable improvement. Parkinsonism actually improved with ECT (mean Hoehn-Yahr stage 1.8). Adverse events: delirium in 24%, amnesia in 4% — all temporary. **No exacerbation of cognitive function before vs after ECT.** The authors did caution that ECT should be performed with caution in patients with severe cognitive impairment or severe brain atrophy.
- **Kobayashi 2026** (n=40 LBD patients vs 57 controls): ECT well-tolerated with no serious adverse events; transient amnesia the most common side effect. Short-term efficacy in LBD comparable to schizophrenia, though lower than primary affective disorders. 2-year psychiatric readmission rate highest in LBD (61.5%) — durability is a real concern.
- **Takamiya 2021** (systematic review/meta-analysis of ECT in Parkinson's, 14 studies, n=129): ECT significantly improved both motor and non-motor symptoms *without worsening cognitive functioning*.
- **No published cases document ECT precipitating motor deterioration in prodromal-synucleinopathy patients.** The available evidence suggests ECT may actually transiently improve parkinsonism, particularly at lower Hoehn-Yahr stages.

BUT THE CUMULATIVE GA EXPOSURE LITERATURE IS REAL AND APPLIES HERE

- **Schulte 2018 (Mayo Clinic Study of Aging, n=1,819, age ≥50):** A single new GA+surgery exposure was associated with an additional 0.2 SD cognitive decline over 5 years beyond normal aging — driven by memory and executive function. The authors specifically noted this "small cognitive decline could be meaningful for individuals with already low baseline cognition."
- **Pennings 2025 (MAAS cohort, n=1,823, 12-year follow-up):** Cumulative total time under GA independently and negatively affected executive functioning, selective attention, and information processing speed. The dose-response runs on cumulative anesthesia minutes, not just exposure count.
- **Linassi 2022:** Aging × prior GA exposure produces OR 9.74 for executive function impairment in older patients with at least one prior GA.
- **ECT-GA is briefer than surgical GA** (~5–10 minutes per session, TIVA only, no surgical inflammation). Per-exposure risk is likely lower than surgical GA. But the cumulative effect of 20–40 brief exposures over an induction + maintenance course in a vulnerable brain is not characterized.

ACUTE DETERIORATION AFTER A SINGLE GA EXPOSURE IS A MEANINGFUL SIGNAL

- **Kunicki 2023 (SAGES cohort, n=560, 6-year follow-up):** Postoperative delirium associated with a 40% acceleration in the slope of cognitive decline out to 72 months — approximating the trajectory seen in patients diagnosed with Alzheimer's within 5 years. The conceptual framework: "*the development of delirium signals underlying brain vulnerability*."
- **Kim 2018 (prospective, age 71.7 ± 4.7, spinal surgery):** RBD was a significant independent predictor of postoperative delirium (OR 1.45). The authors explicitly hypothesized that "underlying alpha synucleinopathy may correlate with postoperative delirium." Dad has formal RBD of 8.5 years duration.
- **The temporal pattern in Dad's case** (onset ~1–2 weeks post-GA, progressive over 5–8 weeks, not fully resolved at 8+ weeks) is consistent with either POCD or GA-triggered acceleration of prodromal neurodegeneration. Either interpretation argues for caution with additional GA exposures.
- **No published study directly examines whether patients who declined after one GA are at higher risk on subsequent exposures.** This is a critical evidence gap. But the conceptual framework strongly implies elevated risk.

Why the ketamine bridge + neurology workup should come first

THE COMPARATIVE-EFFECTIVENESS CASE

- **ELEKT-D (Anand 2023, NEJM, n=403):** Ketamine non-inferior to ECT for nonpsychotic TRD. Response 55.4% (ketamine) vs 41.2% (ECT) — numerically favoring ketamine, statistically non-inferior. Delayed-recall T-score change: **-0.9 with ketamine vs -9.7 with ECT** at end of treatment — substantial cognitive-safety advantage for ketamine.
- **Kumpf 2025 (ELEKT-D secondary analysis):** ECT recipients performed significantly worse on all cognitive tasks at end of treatment; differences resolved among responders at 1-, 3-, and 6-month follow-up. The cognitive cost is transient in responders, but real during and immediately after the treatment course.
- **Menon 2023 meta-analysis** explicitly recommended "a trial of ketamine before a trial of ECT for patients with MDE."
- **Dad's documented responder status** means the prior probability of ketamine working again is meaningfully elevated relative to a ketamine-naïve patient. February 2026: dramatic ~4-week response. The single Royal Oak session 5/18 was difficult, but is not a fairly-tested ketamine course.

THE URGENCY QUESTION — DOES DAD MEET THE THRESHOLD FOR BYPASSING THE WORKUP?

- **Published thresholds for ECT urgency** (per APA / Espinoza & Kellner 2022 NEJM / Unützer 2012 JAMA): active suicidality, catatonia, psychotic features, refusal to eat, life-threatening medical complications of depression.
- **Dad does not currently meet these.** Severe TRD (PHQ-9 = 24) with significant distress, but not suicidal, not catatonic, not psychotic, eating with support, no acute medical complications.
- **What would shift the urgency calculus:** emergence of suicidal ideation, food refusal, catatonic features, rapid functional decline. If any of these emerge, the urgency threshold for ECT would be met regardless of the pending neurology workup, and proceeding would be appropriate with enhanced monitoring.

WHAT "COMPLETING THE WORKUP" ACTUALLY MEANS IN THIS CASE

1. **DAT-SPECT.** The most informative single test. If abnormal: confirms Dad in the Morikawa 2024 patient population for whom ECT safety data exists; strengthens the case for minimizing cumulative GA; informs ECT technique (right unilateral ultrabrief preferred). If normal: substantially attenuates the prodromal-synucleinopathy concern and shifts the risk-benefit more favorably toward ECT.
2. **Structured motor exam (UPDRS or equivalent).** Establishes baseline for monitoring during any ECT course.
3. **Formal cognitive baseline (MoCA + Trail Making Test B at minimum).** Pre-ECT cognitive assessment is the standard of care; serial monitoring during the course depends on having a baseline.
4. **Brain imaging if not already done.** Brain atrophy assessment matters because the Morikawa 2024 cautions specifically named severe brain atrophy as a relative contraindication.

IF ECT DOES PROCEED, WHAT TO ASK FOR

- **Right unilateral ultrabrief pulse** electrode placement — minimizes cognitive side effects and bradycardia risk per Stewart 2011.
- **Propofol-based anesthesia** — lower inflammatory profile (IL-6, TNF- α) and faster cognitive recovery vs methohexital/barbiturate in elderly per Pang 2021.
- **Pre-ECT cardiology clearance** given LAFB + bradycardia; glycopyrrolate availability for ECT-induced bradycardia.
- **Processed EEG monitoring** during anesthesia to avoid excessive anesthetic depth.
- **Serial cognitive monitoring every 3–4 sessions** with the same instruments used at baseline.
- **Low threshold to discontinue** if new motor or cognitive findings or delirium emerge between sessions.

- **Explicit conversation about maintenance ECT timing and total exposure plan** upfront — not after induction is complete.

Key studies supporting the position

ECT IN LEWY-SPECTRUM DISEASE — THE MORE PERMISSIVE EVIDENCE

Morikawa F, Kobayashi R, Murayama T et al. Evaluating Electroconvulsive Therapy for Dementia With Lewy Bodies, Including the Prodromal Stage.

Int J Geriatr Psychiatry 2024;39:e70020.

KEY FINDING Retrospective ECT analysis in 25 DLB / prodromal DLB / MCI-LB patients (all with abnormal DAT-SPECT). 96% at least minimal improvement; 60% remarkable improvement. Parkinsonism improved with ECT. No cognitive deterioration. Cautions about severe brain atrophy. This is the most permissive published data point on ECT in this patient population.

Takamiya A et al. Electroconvulsive Therapy for Parkinson's Disease: Systematic Review and Meta-Analysis.

Movement Disorders 2021;36:50–58.

KEY FINDING 14 studies, n=129 PD patients. ECT significantly improved both motor and non-motor symptoms without worsening cognitive functioning.

KETAMINE NON-INFERIOR TO ECT WITH COGNITIVE ADVANTAGE

Anand A et al. Ketamine vs ECT for Nonpsychotic Treatment-Resistant Major Depression (ELEKT-D).

N Engl J Med 2023;388:2315–2325.

KEY FINDING n=403 nonpsychotic TRD outpatients. Ketamine non-inferior to ECT on response (55.4% vs 41.2%). Delayed-recall T-score change –0.9 (ketamine) vs –9.7 (ECT) — substantial cognitive-safety advantage for ketamine.

CUMULATIVE GA EXPOSURE AND COGNITIVE TRAJECTORY

Schulte PJ et al. Association Between Anaesthesia and Surgery and Long-Term Cognitive Trajectories in Older Adults (Mayo Clinic Study of Aging).

Br J Anaesth 2018;121:398–405.

KEY FINDING n=1,819 adults ≥50. Single new GA+surgery exposure → additional 0.2 SD cognitive decline over 5 years beyond normal aging. Magnified in patients with already-low baseline cognition.

Pennings CH et al. Anaesthesia as a Risk Factor for Long-Term Cognitive Decline (MAAS Cohort).

Eur J Anaesthesiol 2025;42:468–477.

KEY FINDING n=1,823, 12-year follow-up. Cumulative total time under GA independently and negatively affected executive functioning, selective attention, and information processing speed. Dose-response on cumulative anesthesia minutes.

Linassi F et al. Influence of Previous General Anesthesia on Cognitive Impairment.

Front Hum Neurosci 2022;16:810046.

KEY FINDING Aging × prior GA exposure OR 9.74 for executive function impairment in older patients with at least one prior GA.

ACUTE POST-GA DECLINE AS A VULNERABILITY SIGNAL

Kunicki ZJ et al. Six-Year Cognitive Trajectory in Older Adults Following Major Surgery and Delirium (SAGES Cohort).

JAMA Intern Med 2023;183:442–450.

KEY FINDING n=560, 6-year follow-up. Postoperative delirium associated with 40% acceleration in cognitive decline slope — approximating the trajectory of patients diagnosed with Alzheimer's within 5 years. "Delirium signals underlying brain vulnerability."

Kim KH et al. Parkinson's Disease-Related Non-Motor Features as Risk Factors for Post-Operative Delirium in Spinal Surgery.

PLoS One 2018;13:e0195749.

KEY FINDING Prospective study, n=104 elderly. RBD was significant independent predictor of postoperative delirium (OR 1.45). Authors hypothesized "underlying alpha synucleinopathy may correlate with postoperative delirium."

AFTER THE U-M CONSULTATION — WHAT TO CAPTURE IN WRITING

1. What did the U-M ECT team recommend, and what did they say about waiting for the neurology workup?
2. If they recommended proceeding, what was their stated indication? What specific clinical findings would they reverse on?
3. What electrode placement, pulse width, and anesthesia protocol would they use?
4. What's their plan for cognitive baseline and serial monitoring?

5. What's their position on coordination with the U-M Movement Disorders team and the A2 ketamine clinic?
6. If ECT is going to proceed, when, and what's the maintenance plan?